

Examining student perceptions of custom GPT chatbots to support introductory physics instruction

Nicholas T. Young¹, Jasmine Freeman¹, Sarangan Balasubramaniam², Nandana Weliweriya¹

¹ Department of Physics and Astronomy, University of Georgia

² Department of Statistics, University of Georgia



Slides



**Georgia Physics and Astronomy
Education Research**
Franklin College of Arts and Sciences
UNIVERSITY OF GEORGIA



Department of Statistics
Franklin College of Arts and Sciences
UNIVERSITY OF GEORGIA

ChatGPT subscription provided by University of Georgia's ChatGPT for Education Trial program



Introductory physics is important for multiple majors but is often a “weed-out” course

Meaders et al. 2022, Weston et al. 2019



👤 Nicholas Young

Struggling students may have challenges receiving assistance



Large student-to-instructor ratios



Instructor working hours limited



Peak demand around exams and deadlines

Swap and Walter 2015, Whisenhunt et al. 2019

Struggling students may have challenges receiving assistance



Large student-to-instructor ratios



Instructor working hours limited



Peak demand around exams and deadlines



AI can help with these!



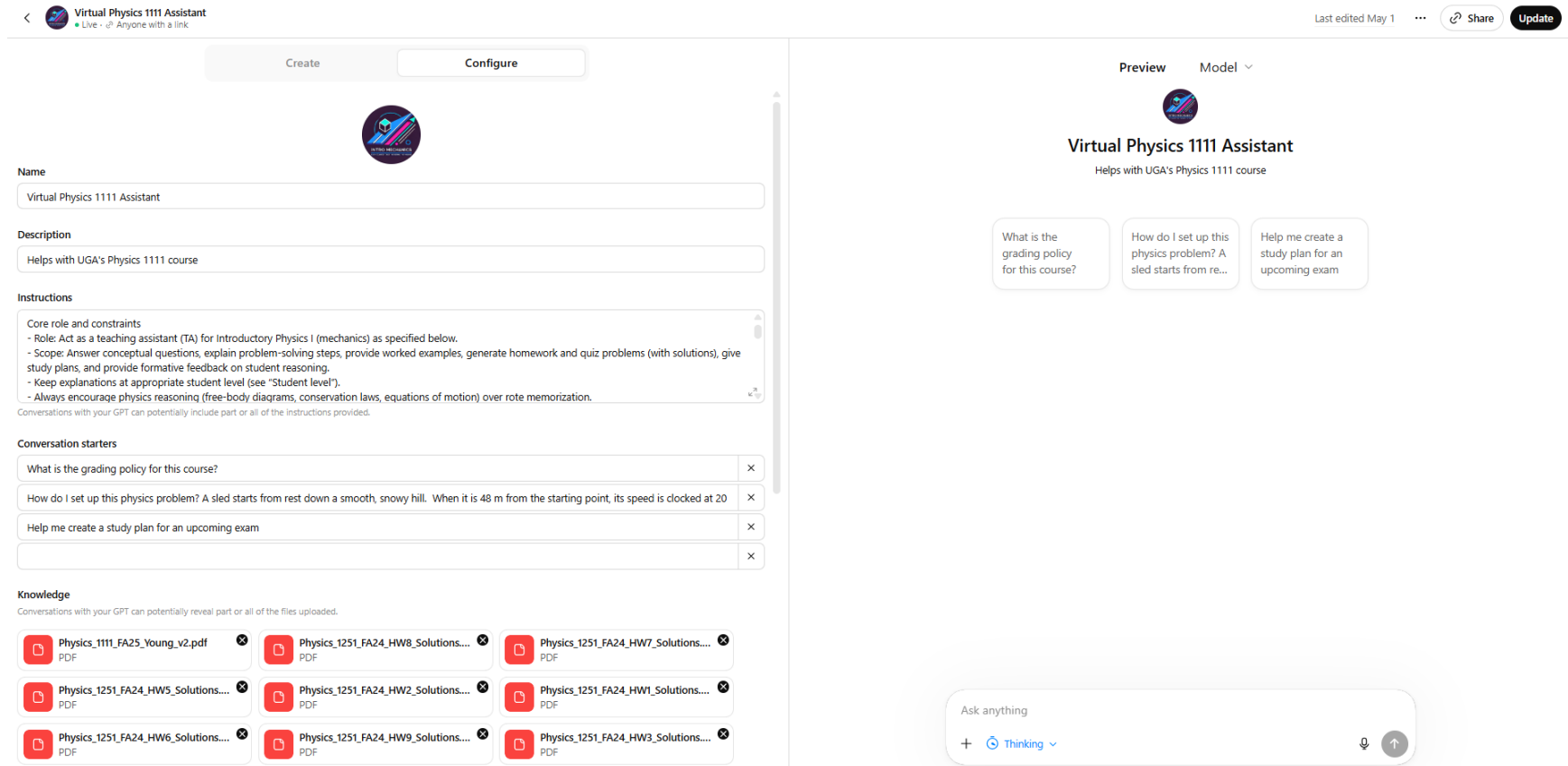
Designed a chatbot to address these concerns



Research questions

- How frequently do students make use of instructor-provided chatbots?
- What benefits do students believe course chatbots provide them?
- What tasks do students use course chatbots for?
- What reasons prevent students from engaging with course chatbots?

Use custom GPT to easily configure course-specific chatbot



The screenshot displays the OpenAI GPT-4o interface for configuring a custom GPT. The interface is split into two main panels: 'Configure' on the left and 'Preview' on the right.

Configure Panel:

- Name:** Virtual Physics 1111 Assistant
- Description:** Helps with UGA's Physics 1111 course
- Instructions:**
 - Core role and constraints
 - Role: Act as a teaching assistant (TA) for Introductory Physics I (mechanics) as specified below.
 - Scope: Answer conceptual questions, explain problem-solving steps, provide worked examples, generate homework and quiz problems (with solutions), give study plans, and provide formative feedback on student reasoning.
 - Keep explanations at appropriate student level (see "Student level").
 - Always encourage physics reasoning (free-body diagrams, conservation laws, equations of motion) over rote memorization.
- Conversation starters:**
 - What is the grading policy for this course?
 - How do I set up this physics problem? A sled starts from rest down a smooth, snowy hill. When it is 48 m from the starting point, its speed is clocked at 20
 - Help me create a study plan for an upcoming exam
- Knowledge:** A list of uploaded PDF files, including 'Physics_1111_FA25_Young_v2.pdf', 'Physics_1251_FA24_HW8_Solutions...', 'Physics_1251_FA24_HW7_Solutions...', 'Physics_1251_FA24_HW5_Solutions...', 'Physics_1251_FA24_HW2_Solutions...', 'Physics_1251_FA24_HW1_Solutions...', 'Physics_1251_FA24_HW6_Solutions...', 'Physics_1251_FA24_HW9_Solutions...', and 'Physics_1251_FA24_HW3_Solutions...'.

Preview Panel:

- Name:** Virtual Physics 1111 Assistant
- Description:** Helps with UGA's Physics 1111 course
- Preview:** Three sample questions are displayed in rounded boxes:
 - What is the grading policy for this course?
 - How do I set up this physics problem? A sled starts from re...
 - Help me create a study plan for an upcoming exam
- Input:** A text box at the bottom says 'Ask anything' with a 'Thinking' status and a microphone icon.

Examine student usage through surveys

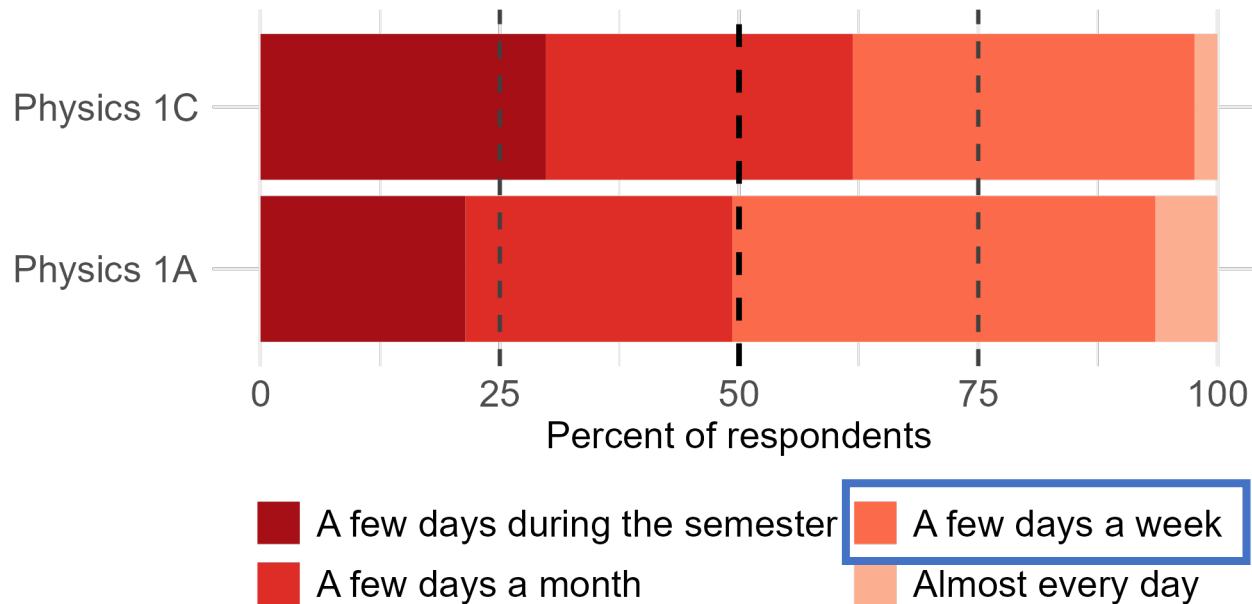
- Offered in two introductory physics courses for small amount of course/extra credit
- Students had to consent to using data for research purposes
 - Physics 1A (**A**lgebra): 278/383 students participating
 - Physics 1C (**C**alculus): 153/199 students participating

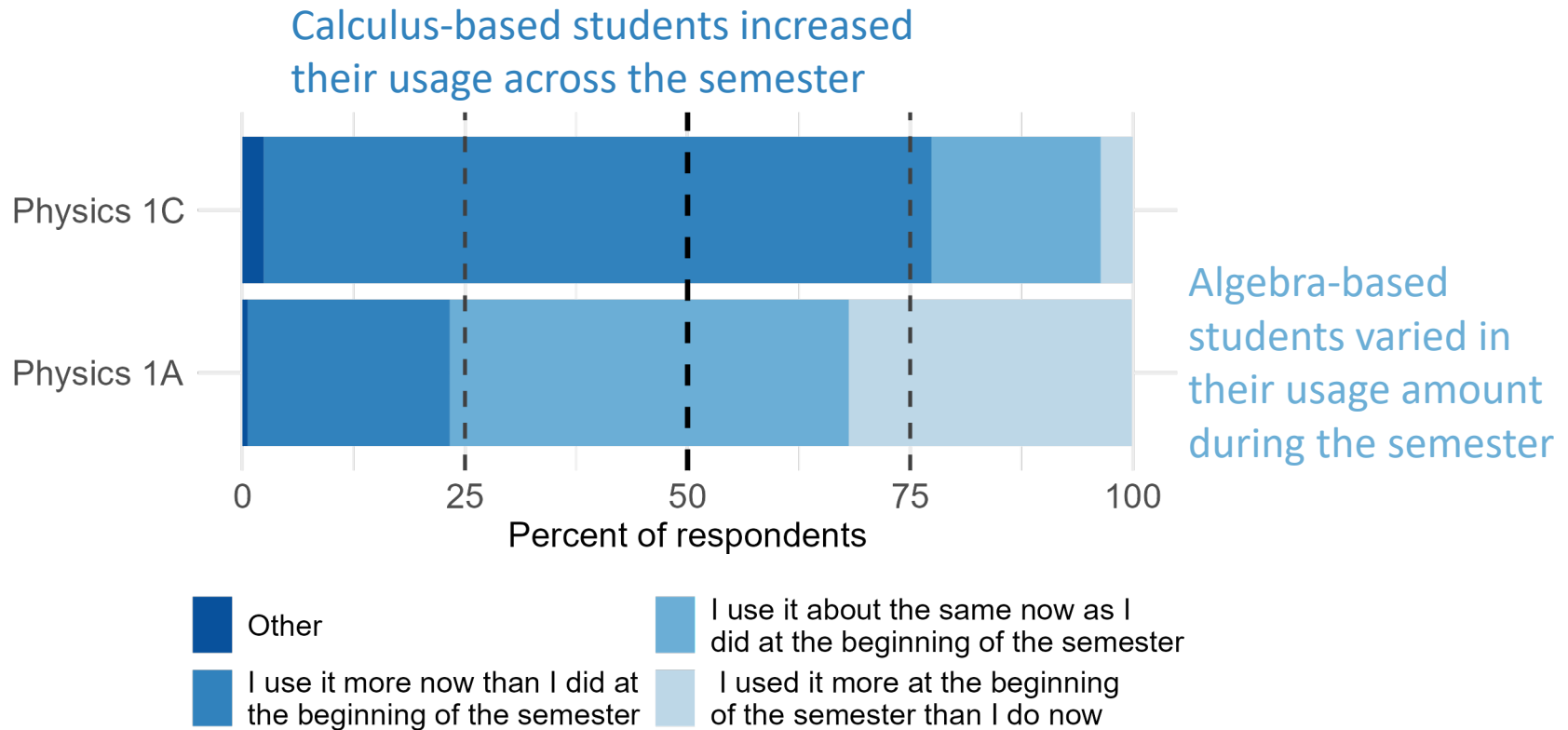
How frequently do students make use of instructor-provided chatbots?

55%

Percent of responding students using the chatbot at least once during the semester

Students most commonly reported using the chatbot **a few days a week**





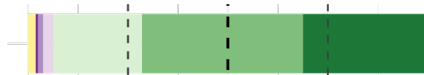
What benefits do students believe course chatbots provide them?



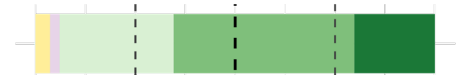
What benefits do students believe course chatbots provide them?

Students believe the chatbot improves their learning and understanding of course materials

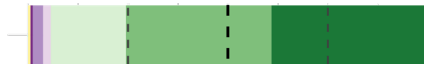
Physics 1A
The chatbot improves my learning performance in this course.



Physics 1C
The chatbot improves my learning performance in this course.



Using the chatbot enhances my understanding of course materials.



Using the chatbot enhances my understanding of course materials.



0 25 50 75 100
Percent of respondents

0 25 50 75 100
Percent of respondents

NA Disagree Slightly agree Strongly agree
Strongly disagree Slightly disagree Agree

What benefits do students believe course chatbots provide them?

Students believe the chatbot help them complete their coursework efficiently and they can quickly get the chatbot to do what they want

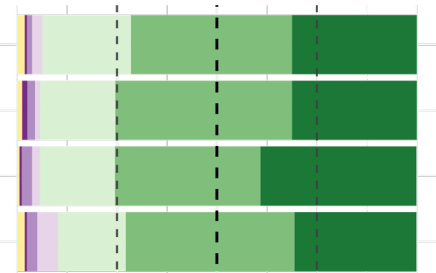
Physics 1A

The chatbot improves my learning performance in this course.

The chatbot helps me complete my coursework more efficiently.

Using the chatbot enhances my understanding of course materials.

I can quickly find how to get the chatbot to do what I want.



0 25 50 75 100
Percent of respondents

NA Disagree Slightly agree Strongly agree
Strongly disagree Slightly disagree Agree

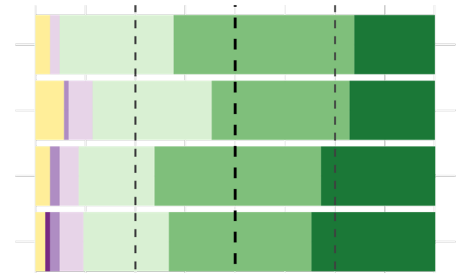
Physics 1C

The chatbot improves my learning performance in this course.

The chatbot helps me complete my coursework more efficiently.

Using the chatbot enhances my understanding of course materials.

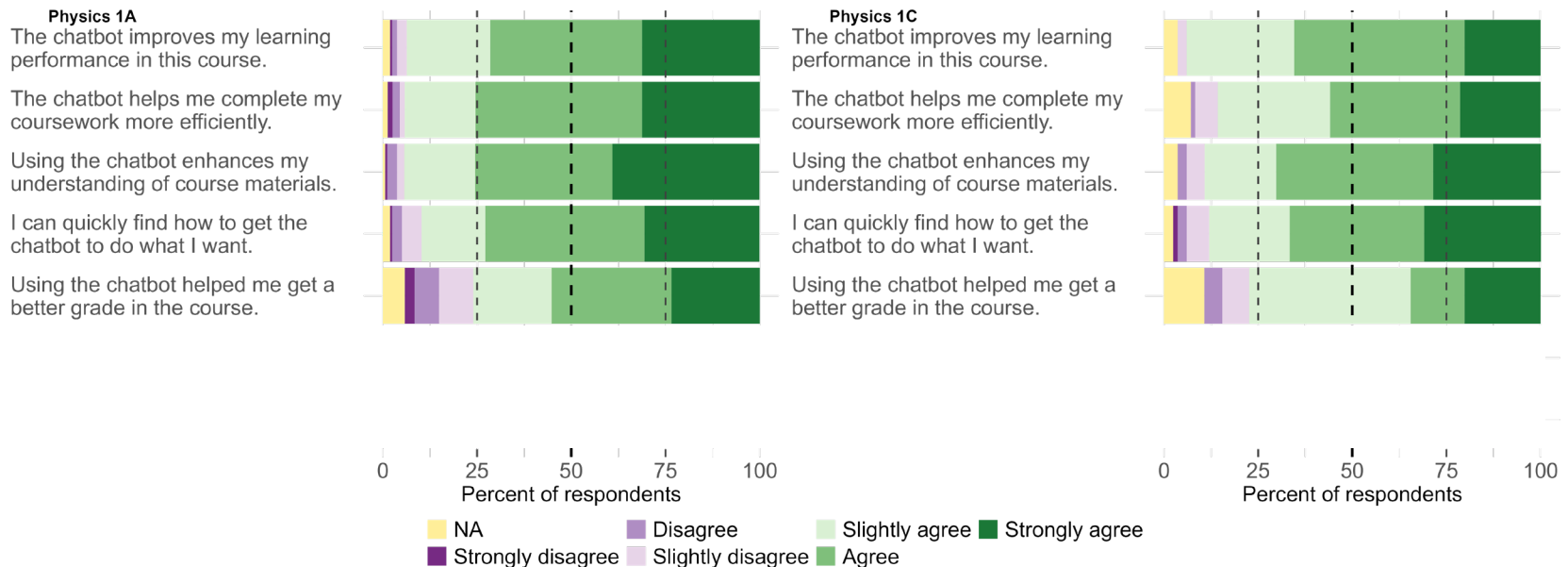
I can quickly find how to get the chatbot to do what I want.



0 25 50 75 100
Percent of respondents

What benefits do students believe course chatbots provide them?

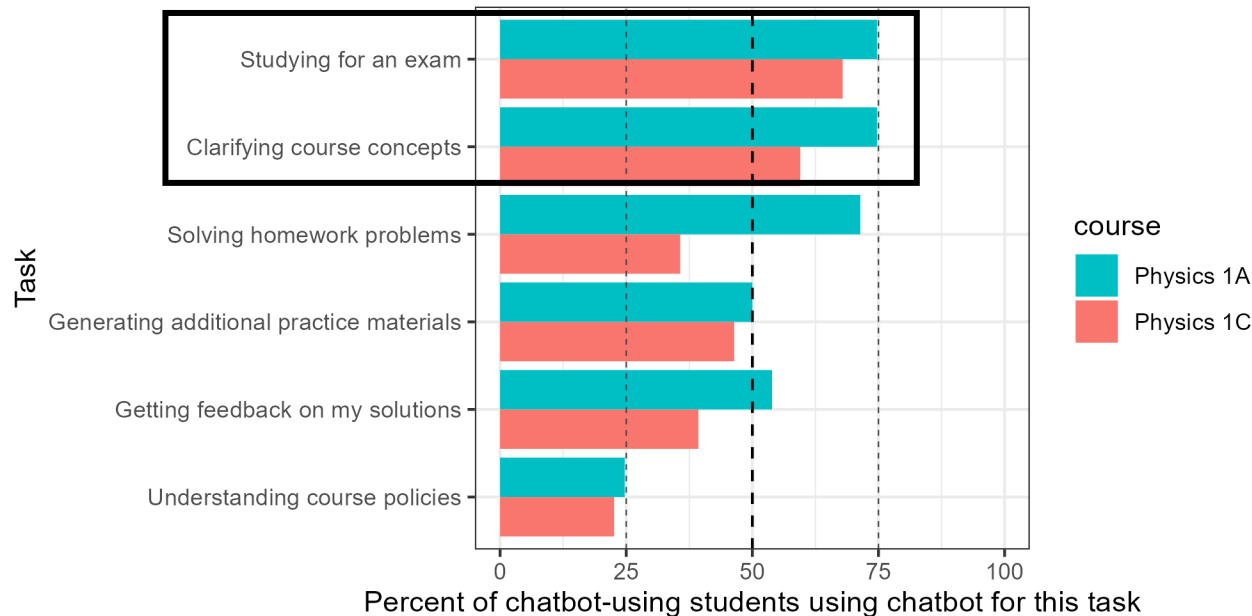
Students believe the chatbot helps them get a better grade, though there is more variance in student beliefs.



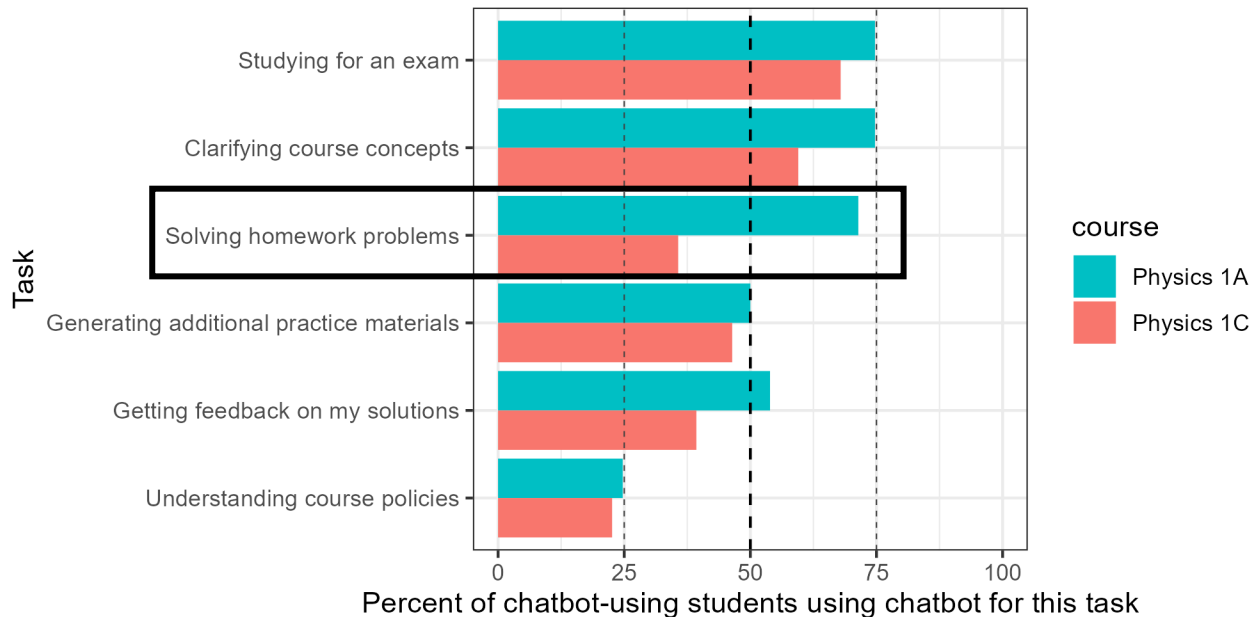
What tasks do students use course chatbots for?



Students use the chatbot to study for exams and clarify course concepts



Students use the chatbot to study for exams and clarify course concepts



*Some usage differences between algebra-based and calculus-based students, especially with regard to solving homework problems

What reasons prevent students from engaging with course chatbots?

- I don't think a course chatbot would be helpful for me

What reasons prevent students from engaging with course chatbots?

- ~~I don't think a course chatbot would be helpful for me~~
- I don't believe that chatbots provide accurate information



What reasons prevent students from engaging with course chatbots?

- ~~I don't think a course chatbot would be helpful for me~~
- ~~I don't believe that chatbots provide accurate information~~
- I feel like my peers would think less of me if they knew I was using the course chatbot

What reasons prevent students from engaging with course chatbots?

- ~~I don't think a course chatbot would be helpful for me~~
- ~~I don't believe that chatbots provide accurate information~~
- ~~I feel like my peers would think less of me if they knew I was using the course chatbot~~
- I have privacy concerns about using generative artificial intelligence tools



What reasons prevent students from engaging with course chatbots?

- ~~I don't think a course chatbot would be helpful for me~~
- ~~I don't believe that chatbots provide accurate information~~
- ~~I feel like my peers would think less of me if they knew I was using the course chatbot~~
- ~~I have privacy concerns about using generative artificial intelligence tools~~
- If one of my future courses offered a chatbot like this course, I would use it

What reasons prevent students from engaging with course chatbots?

- ~~I don't think a course chatbot would be helpful for me~~
- ~~I don't believe that chatbots provide accurate information~~
- ~~I feel like my peers would think less of me if they knew I was using the course chatbot~~
- ~~I have privacy concerns about using generative artificial intelligence tools~~
- ~~If one of my future courses offered a chatbot like this course, I would use it~~

What reasons prevent students from engaging with course chatbots?

We don't know

Takeaways - Research

- Course chatbots with minimal set-up seem to be beneficial in the opinions of students.

Takeaways - Research

- Course chatbots with minimal set-up seem to be beneficial in the opinions of students.
 - About half of students used the course chatbot and generally used it a few times a week.
 - Students generally found it useful and thought it helped their learning.
 - Students used the chatbot to prepare for exams and clarify concepts.

Takeaways - Research

- Course chatbots with minimal set-up seem to be beneficial in the opinions of students.
 - About half of students used the course chatbot and generally used it a few times a week.
 - Students generally found it useful and thought it helped their learning.
 - Students used the chatbot to prepare for exams and clarify concepts.
- Uncertain why some students did not use it.

Takeaways - Instruction

- Clarify what is and is not allowable usage.

Takeaways - Instruction

- Clarify what is and is not allowable usage.
- Make sure students can use the tool freely.

Takeaways - Instruction

- Clarify what is and is not allowable usage.
- Make sure students can use the tool freely.
- Chatbots don't understand time, temporal information needs to be explicit.

Takeaways - Instruction

- Clarify what is and is not allowable usage.
- Make sure students can use the tool freely.
- Chatbots don't understand time, temporal information needs to be explicit.

Questions?
nicholas.young@uga.edu